WEEK 5 & 6: RELATIONAL DATABASE DESIGN

Phase 3: Relational Transformation and Practical Activities

MASTERING THE DATABASE DESIGN LIFECYCLE

Bridge the gap between business requirements and physical implementation.

- **Holistic View**: Understand how a real-world problem evolves into a structured database.
- **Phase 3 Focus**: Transitioning from conceptual ER models to logical relational schemas.
- **Goal**: Develop the ability to create systematic, error-free transformations.
- **Practical Application**: Engage in hands-on activities across diverse business scenarios.

CONCEPT EXAMPLE:

Transforming a "Customer" entity from an ERD into a "Customers" table with a defined Primary Key (CustomerID).

THE FOUR-STEP TEACHING PROGRESSION

A logical path from business needs to a functional database.

1 **BUSINESS REQUIREMENT**

Identify core problems.

Example: "A hotel needs to track room availability and guest bookings."

2 **ER DIAGRAM (CONCEPTUAL)**

Map out entities and relationships.

Example: Defining 'Guest' and 'Room' entities with a 'Books' relationship.

3 **RELATIONAL TABLES (LOGICAL)**

Transform into tables with keys.

Example: Creating 'Guests' and 'Rooms' tables with Primary Keys.

4 **DATABASE (PHYSICAL)**

Implement design using SQL DDL.

Example: Running 'CREATE TABLE Guests (...);' in a DBMS.

TRANSITIONING FROM CONCEPTUAL TO LOGICAL

Translating business rules into structural integrity.

- **Entity to Table**: Each entity in your ERD becomes a table in your relational schema.
- **Attribute Mapping**: Attributes become columns, with specific data types and constraints.
- **Relationship Resolution**: 1:N relationships use foreign keys; M:N require junction tables.
- **Integrity Constraints**: Define Primary Keys (PK) for uniqueness and Foreign Keys (FK) for referential integrity.

MAPPING EXAMPLE (1:N):

A "Department" (1) has many "Employees" (N). The DepartmentID is added as a Foreign Key to the Employees table.

STUDENT ENROLLMENT SYSTEM (CONCEPTUAL)

Scenario: A university manages student information, courses offered, and the enrollment links between them.

ENTITIES & ATTRIBUTES

- **Student**: StudentID (PK), Name
- **Course**: CourseID (PK), CourseName

RELATIONSHIP

Many-to-Many (M:N)

Students enroll in multiple courses; courses host multiple students.

EXAMPLE: STUDENT ENROLLMENT SYSTEM

Converting the ERD into a set of relational tables.

- Resolve M:N relationships via junction tables.
- Enforce referential integrity with Foreign Keys.

Student

(**<u>StudentID</u>**, Name)

Course

(**<u>CourseID</u>**, CourseName)

Enrollment

(**<u>EnrollmentID</u>**, **StudentID**, **CourseID**, EnrollmentDate)

- StudentID: FK referencing Student
- CourseID: FK referencing Course

ACTIVITY: ERD TO RELATIONAL TABLE CONVERSION

Objective: Practice systematic conversion from conceptual ERDs to logical relational schemas.

INSTRUCTIONS & TASKS

- **Task 1:** Create a clear ER Diagram using Crow's Foot notation.
- **Task 2:** Convert the ERD into Relational Tables with PKs and FKs.
- **Deliverables:** Specify table names, columns, and constraints.

SAMPLE DELIVERABLE FORMAT:

```
Table: Students
Columns: StudentID (PK), Name, MajorID (FK)
```

SCENARIO 1: HOSPITAL SYSTEM

CORE ENTITIES

- **Patient**: Personal records and admissions
- **Doctor**: Medical staff and specialists
- **Appointment**: Scheduled consultations
- **Treatment**: Medical procedures
- **Ward & Bed**: Facility infrastructure

RELATIONSHIP MAPPING

- **Patient-Ward**: 1:N (One ward houses many patients)
- **Doctor-Patient**: 1:N via Appointment (One doctor, many patients)
- **Patient-Treatment**: M:N (Patients receive multiple treatments)

Critical Challenge:

How will you handle the M:N relationship between Doctors and Treatments to ensure full traceability over time?

SCENARIO 2: HOTEL RESERVATION SYSTEM

CORE ENTITIES

- **Guest**: Personal information and history
- **Room**: Physical inventory and status
- **Reservation**: Booking details and dates
- **RoomType**: Categories (Standard, Deluxe, Suite)
- **Staff**: Employees managing reservations

KEY CONSTRAINTS

- **Room-RoomType**: 1:N (One type, many rooms)
- **Guest-Reservation**: 1:N (One guest, many stays)
- **Staff-Reservation**: 1:N (One staff manages many bookings)

Critical Data Points:

Track GuestID, RoomNumber, Price, and specific Check-In/Out dates to ensure accurate billing and availability.

SCENARIO 3: INVENTORY & SUPPLY CHAIN

CORE ENTITIES

- **Product**: Inventory items and specifications
- **Supplier**: External vendors and sources
- **Order**: Customer purchase records
- **Customer**: Client profiles and history
- **Warehouse**: Storage locations and logistics

COMPLEX RELATIONSHIPS

- **Product-Supplier**: M:N (Multiple sources for products)
- **Order-Product**: M:N (One order contains many items)
- **Product-Warehouse**: M:N (Inventory spread across locations)

Mapping Goal:

Ensure zero data redundancy while maintaining full traceability of orders across the entire supply chain.

SUMMARY & BEST PRACTICES

Key takeaways for successful relational transformation.

- **Consistency is Key**: Ensure naming conventions are uniform across all tables.
- **Verify Cardinality**: Double-check 1:N vs M:N relationships before creating tables.
- **Integrity First**: Never skip defining Foreign Keys; they are the backbone of relational data.
- **Documentation**: Keep business requirements updated as the design evolves.